Bangladesh University of Textiles

Department of Fabric Engineering

Subject Name: Fabric Manufacturing - II (Practical)

Subject Code: FE 352-0723

Department: Department of Wet Process Engineering

Lab Teacher: Abdullah Sayam, Lecturer, DoFE

Routine:

Exp No.	Experiment Name
01	Study on the Layout Plan of Knitting Lab
02	Study on the Primary Weft Knitting and Warp Knitting Elements
03	Study on the Knitting Action of Latch Needle and the Conventional Hand Socks Knitting Machine
04	Study on the Different Parts of a Modern 4-Track Single Jersey Circular Knitting Machine
05	Study on Needle Arrangement and Cam Arrangement of a Single Jersey Circular Knitting Machine
06	Study on Rib Circular Knitting Machine
07	Study on Interlock Circular Knitting Machine
08	Study on V-Bed Flat Knitting Machine
09	Study Determination of Stitch Length (SL), Wales per Inch (WPI), Courses per Inch (CPI) from Given Sample and Production Calculation
10	Study on the Tricot Warp Knitting Machine
11	Final Written Exam (20 Marks)
12	Final Viva (20 Marks)

Mark Distribution:

Continuous Internal Evaluation (CIE)			**Term End Examination (TEE)**		**Total**
Attendance	Lab Report	Daily Performance	Final Written Exam	Final Viva	
10	10	40	20	20	100

Note: All students are required to bring a printed copy of the lap report job sheet of the respective experiment during each lab.

□ Excellent	[10]
□ Very Good	[9]
□ Good	[8]
□ Satisfactory	[7]
□ Bad	[6]

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Department of Fabric Engineering

FE 352-0723: Fabric Manufacturing - II (Practical)

Dept: ________________Date: _____________

Name: ID:

Exp No: **01** Name of the Experiment: **Study on the Layout Plan of Knitting Lab**

Sign:	Key words: Ratio factor, scaling, specification, layout objectives, advantages, considering points before layout, machine specifications and outputs.

1. What do you mean by layout plan? List down the different types of layouts used.

2. Write down the objectives of making a layout plan.

3. List down the points to be considered before layout planning?

4. **Draw the layout of the knitting lab with major specifications.**

5. Complete the following chart according to the drawn layout plan.

M/C No.	Machine Name & Type	Brand	Origin	Dia & Gauge (D×G)	No. of Feeders	M/C Output
01						
02						
03						
04						
05						
06						
07						
08						
09						
10						
11						
12						

□ **Excellent**	**[10]**
□ **Very Good**	**[9]**
□ **Good**	**[8]**
□ **Satisfactory**	**[7]**
□ **Bad**	**[6]**

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FE 352-0723: Fabric Manufacturing - II (Practical)

Dept: ____________________Date: _____________

Name: ID:

Exp No: **02** Name of the Experiment: **Study on the Primary Weft Knitting and Warp Knitting Elements**

Sign:	Key words: latch needle, Spring bearded needle, Compound needle, knit cam, tuck cam, miss cam, sinker, tricot guide, tricot sinker & chain link.

1. Draw the primary weft knitting elements with their functions.

1(a) Needle:

Latch Needle	Spring Bearded Needle	Compound Needle
Function of Needle:		

1(b) Sinker:

	Function of Sinker:
Sinker	

1(c) Cam:

Knit Cam	Tuck Cam	Miss Cam
Function of Cam:		

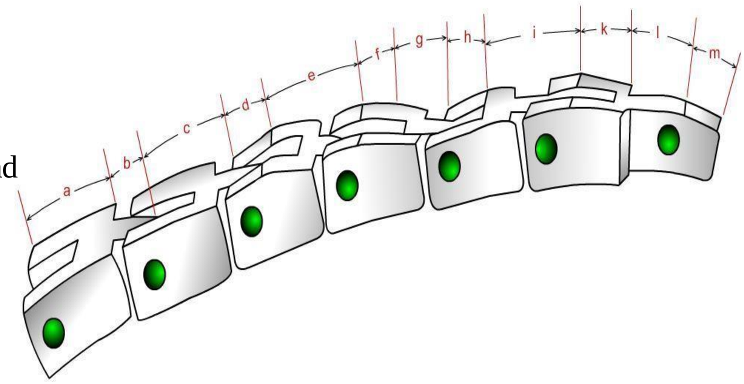
2. Draw the primary warp knitting elements (Tricot Machine) with their functions.

	Function of Tricot Guide:
Tricot Guide	

	Function of Tricot Sinker:
Tricot Sinker	

Tricot Chain Link:

These links are designed by letters indicating the ground edge-
'a'- is an underground link (no ground)
'b'- is a link on which the fork is ground
'c'- is a link on which the tail is ground
'd'- is a link on which both the fork and tail are ground



"a" type link	"b" type link	"c" type link	"d" type link

Function of Chain Link:

☐ Excellent	[10]
☐ Very Good	[9]
☐ Good	[8]
☐ Satisfactory	[7]
☐ Bad	[6]

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Department of Fabric Engineering

FE 352-0723: Fabric Manufacturing - II (Practical)

Dept: ____________________Date: _____________

Name: ID:

Exp No: **03** Name of the Experiment: **Study on the Knitting Action of Latch Needle and the Conventional Hand Socks Knitting Machine**

Sign:	Key words: Knitting action of Single Jersey m/c, mechanism of conventional hand socks m/c with different parts, yarn to fabric path diagram.

1. Show the knitting action of latch needle for producing SJ Knit fabric.

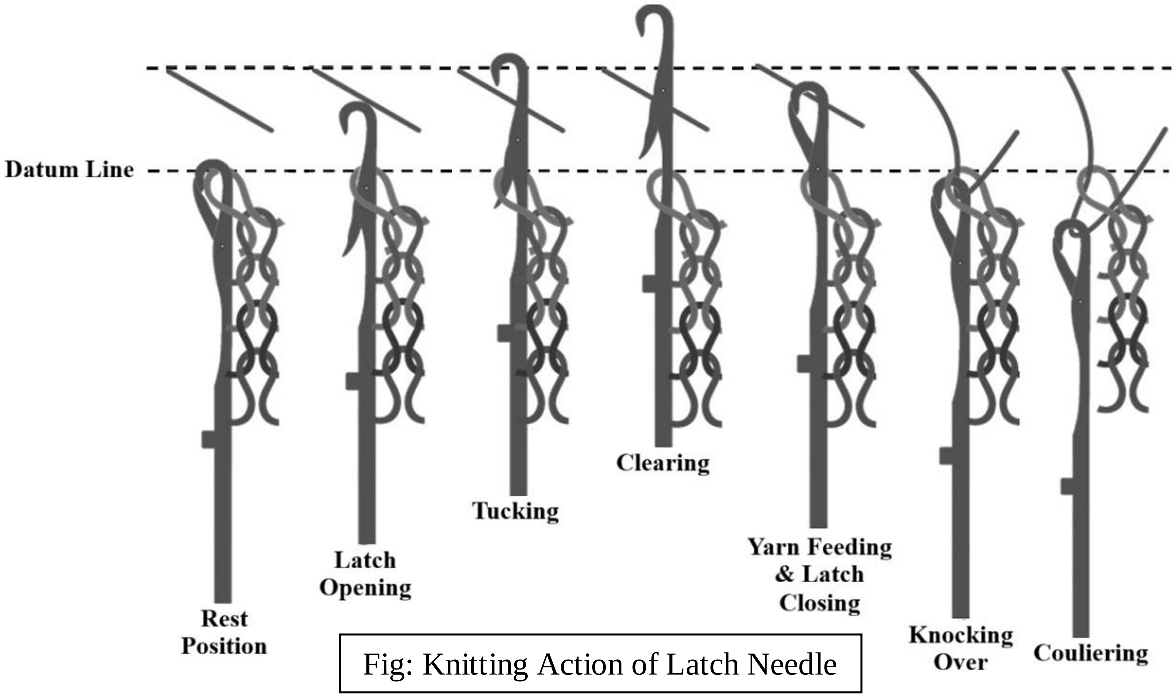

Fig: Knitting Action of Latch Needle

2. **Write short notes on conventional hand socks knitting machine.**

- ☐ Excellent [10]
- ☐ Very Good [9]
- ☐ Good [8]
- ☐ Satisfactory [7]
- ☐ Bad [6]

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FE 352-0723: Fabric Manufacturing - II (Practical)

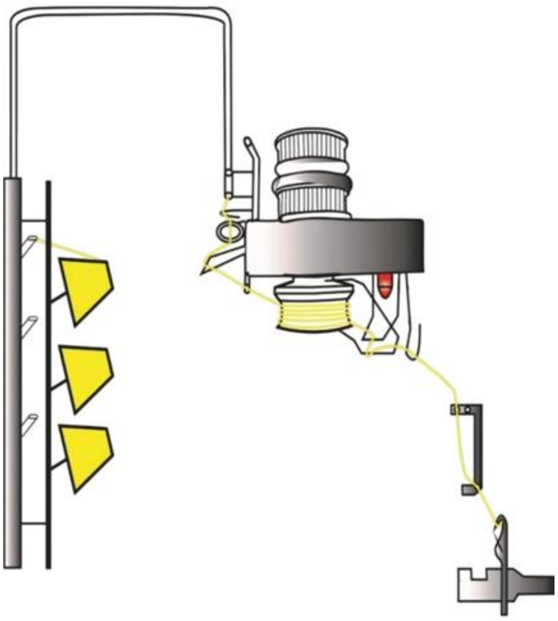
Dept: ____________________Date: _____________

Name: ID:

Exp No: **04** Name of the Experiment: **Study on the Different Parts of a Modern 4-Track Single Jersey Circular Knitting Machine**

Sign:	Key words: creel stand, positive feeder, Lycra feeder, VDQ pulley, belt, spreader, knot catcher, tensioner, guide, spreader, feeder, latch needle, cylinder, cam, cam box, trick, track, gauge, pitch, diameter, takedown roller, cloth roller.

1. Sketch the Yarn Path Diagram of a S/J Knitting Machine and indicate the key parts.



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2. Sketch the positive feed system and mention the functions of the accumulator.

3. Sketch the Lycra feed system. What do you understand by full feeder and half feeder Lycra?

4. What is the purpose of VDQ Pulley in a S/J circular knitting machine?

- ☐ **Excellent** [10]
- ☐ **Very Good** [9]
- ☐ **Good** [8]
- ☐ **Satisfactory** [7]
- ☐ **Bad** [6]

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FE 352-0723: Fabric Manufacturing - II (Practical)

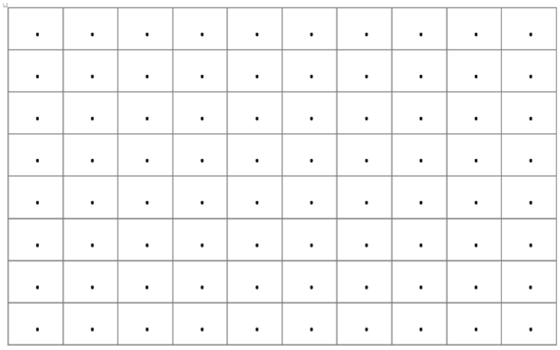
Dept: ________________Date: _____________

Name: ID:

Exp No: **05** Name of the Experiment: **Study on Needle Arrangement and Cam Arrangement of a Single Jersey Circular Knitting Machine**

Sign:	Key words: Notation diagram, needle arrangement, cam arrangement, repeat size, minimum track for specific design, calculation of specific needles & cams, needle and cam calculation.

1. Draw the cam & needle arrangement by using minimum tracks for the given notation diagrams.

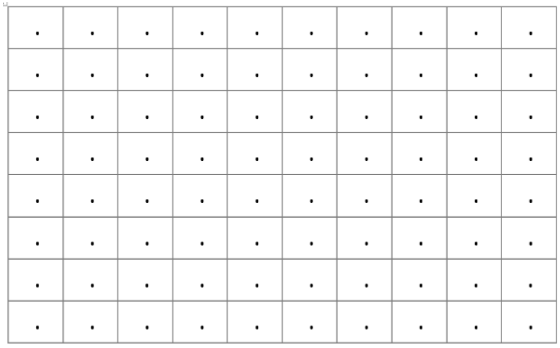


Notation Diagram

Cam Arrangement:

	F1	F2	F3	F4	F5	F6
T1						
T2						
T3						
T4						

Needle Arrangement:

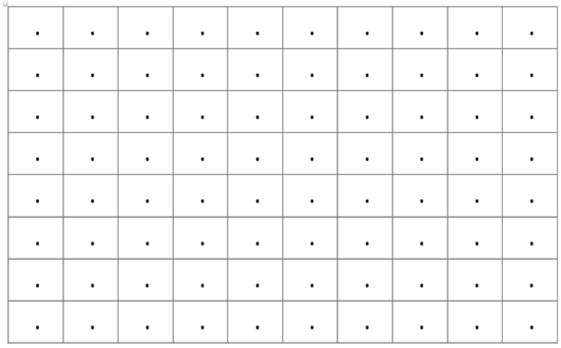


Notation Diagram

Cam Arrangement:

	F1	F2	F3	F4	F5	F6
T1						
T2						
T3						
T4						

Needle Arrangement:



Notation Diagram

Cam Arrangement:

	F1	F2	F3	F4	F5	F6
T1						
T2						
T3						
T4						

Needle Arrangement:

2. **Analyze the SJ fabric sample and draw its notation diagram, cam & needle arrangement.**

Sample Attachment

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Notation Diagram

Cam Arrangement:

	F1	**F2**	**F3**	**F4**	**F5**	**F6**	**F7**	**F8**	**F9**	**F10**
T1										
T2										
T3										
T4										

Needle Arrangement (for min track):

3. **Show the calculation for the number & types of needles and number & types of cams required to produce the fabric on our single jersey 4 track circular knitting machine.**

Needle Calculation	
Butt Position – 1	
Butt Position – 2	
Butt Position – 3	
Butt Position – 4	
Total Needles	

Cam Calculation	
Knit Cams	
Tuck Cams	
Miss Cams	
Total Cams	

- □ Excellent [10]
- □ Very Good [9]
- □ Good [8]
- □ Satisfactory [7]
- □ Bad [6]

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FE 352-0723: Fabric Manufacturing - II (Practical)

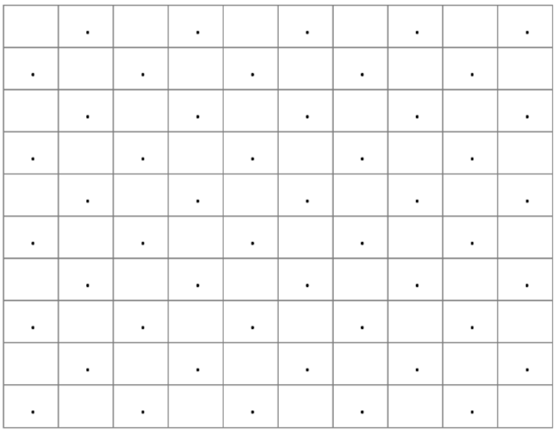
Dept: ____________________Date: _____________

Name: ID:

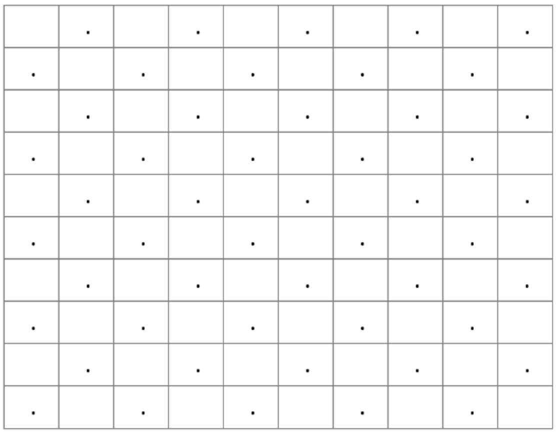
Exp No: **06** Name of the Experiment: **Study on Rib Circular Knitting Machine**

Sign:	Key words: Identification of Rib fabric, cam and needle arrangement, rib gaiting and interlock gaiting, knitting action for 1×1 Rib fabric, differentiate between S/J and Rib machine.

1. Identify the following Rib Fabrics, draw their notation diagrams, cam & needle arrangement.



1×1 Rib



Swiss Pique

Cam & Needle Arrangement:

Dial	F1	F2	F3	F4	Needle
T1					
T2					

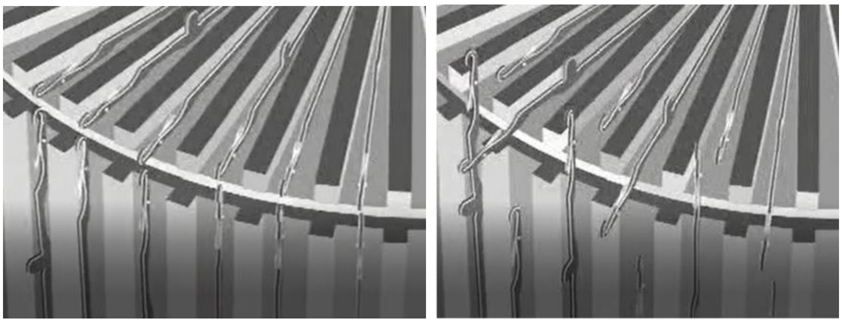
Cylinder	F1	F2	F3	F4	Needle
T1					
T2					
T3					
T4					

Cam & Needle Arrangement:

Dial	F1	F2	F3	F4	Needle
T1					
T2					

Cylinder	F1	F2	F3	F4	Needle
T1					
T2					
T3					
T4					

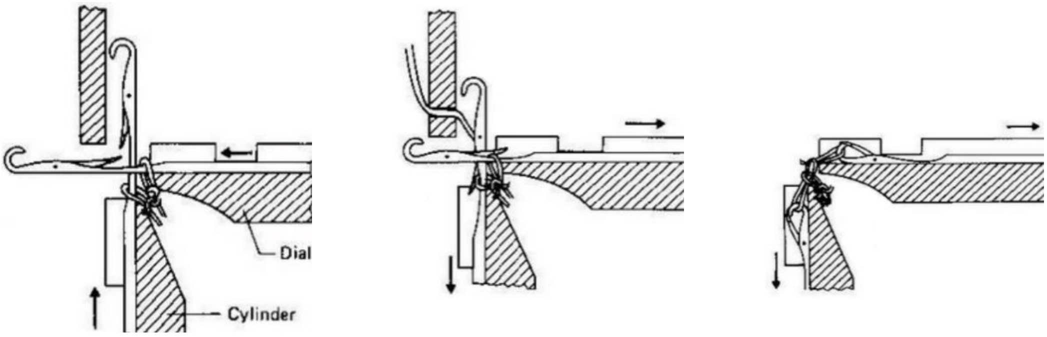
Rib Gaiting



Interlock Gaiting

2. What do you understand by Rib Gaiting & Interlock Gaiting? Why is needle gaiting important in a DJ Circular Knitting Machine?

3. Show the knitting action of latch needle for producing 1×1 Rib fabric in a Rib circular knitting M/C



Clearing Position | Yarn Feeding | Knocking Over

4. How can you differentiate between single jersey and double jersey Rib circular knitting machine?

Single Jersey Circular Knitting Machine	Double Jersey Rib Circular Knitting Machine

□ Excellent	[10]
□ Very Good	[9]
□ Good	[8]
□ Satisfactory	[7]
□ Bad	[6]

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Department of Fabric Engineering

FE 352-0723: Fabric Manufacturing - II (Practical)

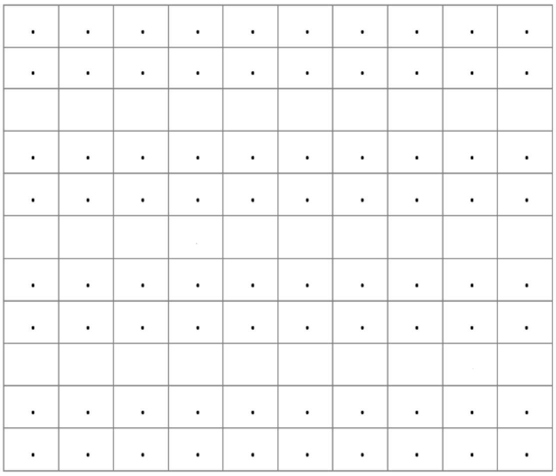
Dept: ____________________Date: _____________

Name: ID:

Exp No: **07** Name of the Experiment: **Study on Interlock Circular Knitting Machine**

Sign:	Key words: Identification of Interlock fabric, cam and needle arrangement, long needle, short needle, interlock gaiting, fabric properties, alternative of long & short needles.

1. Identify the following Interlock Fabrics, draw their notation diagrams, cam & needle arrangement.

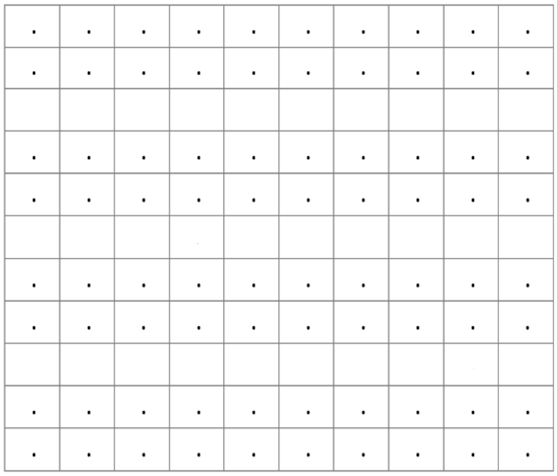


1×1 Interlock

Cam & Needle Arrangement:

Dial	F1	F2	F3	F4	Needle
T1					
T2					

Cylinder	F1	F2	F3	F4	Needle
T1					
T2					
T3					
T4					



Punto De Roma

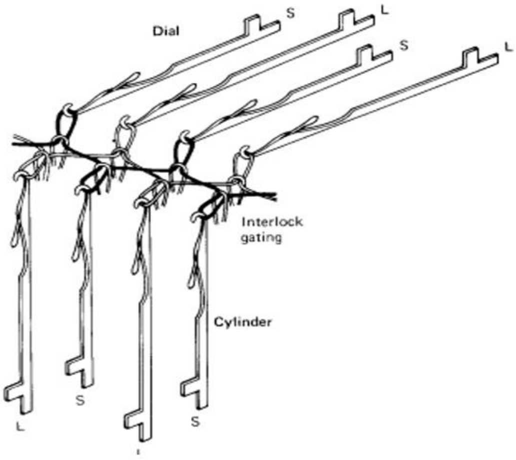
Cam & Needle Arrangement:

Dial	F1	F2	F3	F4	Needle
T1					
T2					

Cylinder	F1	F2	F3	F4	Needle
T1					
T2					
T3					
T4					

2. Write short note on conventional interlock knitting by long needle & short needle.

3. **Sketch the Interlock gaiting of a conventional knitting machine with long needle & short needle.**



4. **How can you differentiate between a Rib and an Interlock circular knitting machine?**

Double Jersey Rib Circular Knitting Machine	Double Jersey Interlock Circular Knitting Machine

5. **Differentiate between the properties of a plain 1×1 Rib and a plain 1×1 Interlock fabric.**

Properties	**Plain 1×1 Rib Fabric**	**Plain 1×1 Interlock Fabric**
Appearance		
Extensibility		
Edge Curling		
Unraveling		
End Uses		

6. **What is the alternative of long & short needles used in modern knitting machines in knit industries.**

- □ Excellent [10]
- □ Very Good [9]
- □ Good [8]
- □ Satisfactory [7]
- □ Bad [6]

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FE 352-0723: Fabric Manufacturing - II (Practical)

Dept: ____________________Date: _____________

Name: ID:

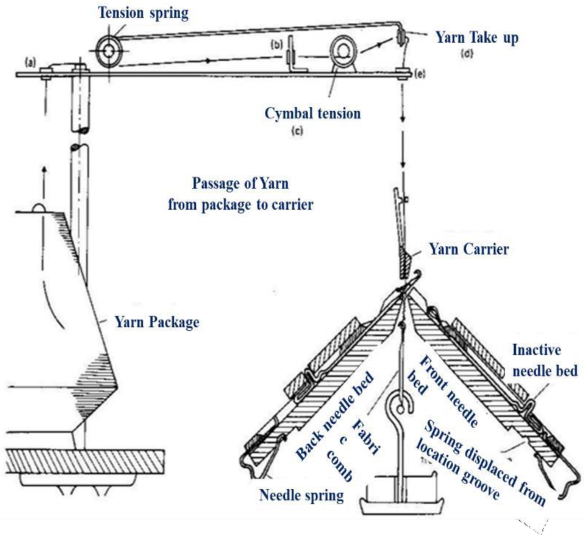
Exp No: **08** Name of the Experiment: **Study on V-Bed Flat Knitting Machine**

Sign:	Key Words: Raising cam, tuck cam, stitch cam, brush, racking liver, cam carriage, Yarn Path diagram, Production system of Half & Full cardigan.

1. **Draw the cam system of a V-bed flat knitting machine of our lab and list down the function of each cam.**

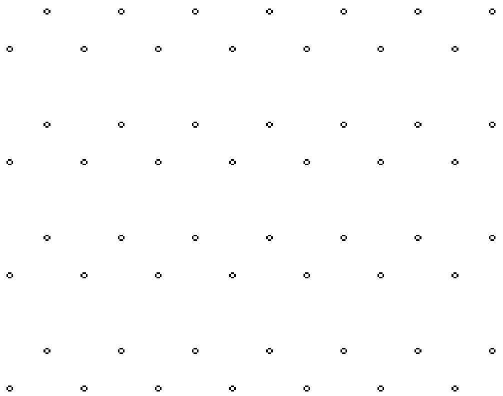
2. **Explain the half in action & full in action function of the Raising Cam.**

3. **Draw the Yarn to fabric path diagram of V-bed flat knitting machine.**

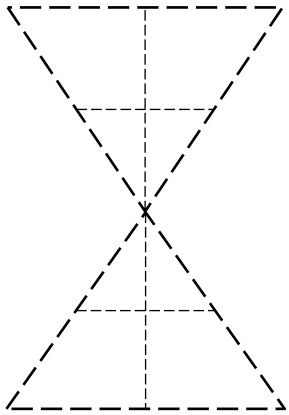


4. What is the function of Brush in the yarn feeding zone?

5. Show Notation diagram, Needle Arrangement and Cam Arrangement of Half Cardigan Fabric.



Notation Diagram



Cam Arrangement

From Left to Right

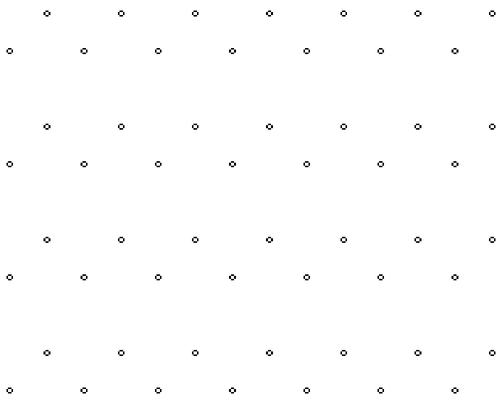
Front Bed:
Back Bed:

From Right to Left

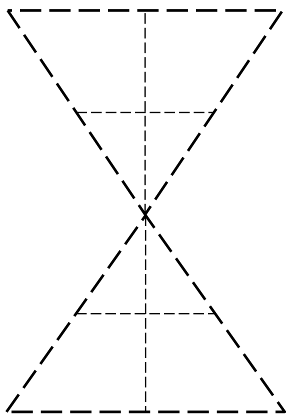
Front Bed:
Back Bed:

Show the Needle Arrangement:

6. Show Notation diagram, Needle Arrangement and Cam Arrangement of Full Cardigan Fabric.



Notation Diagram



Cam Arrangement

From Left to Right

Front Bed:
Back Bed:

From Right to Left

Front Bed:
Back Bed:

Show the Needle Arrangement:

☐ Excellent	[10]
☐ Very Good	[9]
☐ Good	[8]
☐ Satisfactory	[7]
☐ Bad	[6]

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FE 352-0723: Fabric Manufacturing - II (Practical)

Dept: ___________________Date: ______________

Name: ID:

Exp No: **09** Name of the Experiment: **Study Determination of Stitch Length (SL), Wales per Inch (WPI), Courses per Inch (CPI) from Given Sample and Production Calculation**

Sign:	Key words: Stitch length, significance of stitch length, SL, WPI and CPI calculation of single jersey structure, Production calculation.

1. What is stitch length (SL)? How is the SL of a weft knitted fabric measured?

2. Why is stitch length important? What are the affected areas due to change in stitch length?

3. Determination of Stitch Length (SL) from Given Sample.

- Two points are marked (as such that there are 100 loops in between them) at the edge of the fabric from where a full course is unraveled.
- Then a full course is withdrawn from the fabric.
- The course is held at one end by the observer while a certain weight is hanged at the other end to straighten it.
- Then the marked two points are identified on the unraveled course and the value of the distance between these two points is measured with the help of a vertically held ruler.
- The procedure is repeated 4 more times.
- The average of the values is calculated.
- Then the value is divided by 100 (no. of loops) to calculate the stitch length. Generally, it is expressed in millimeter.

No. of Observations	No. of loops	Straightened Yarn Length (cm)	Average of Straightened Yarn Length (cm)	Stitch Length (cm)	Stitch Length (mm)
1	100				
2					
3					

4. Determination of Courses per Inch (CPI) from the Given Sample.

- The courses are unraveled from the cut sample one after another.
- When any of the courses touches the top side of the square, then the counting for CPI should be started.
- The courses will be unraveled one by one until the other side of the square is touched by any of the unraveled courses. Then the counting should be stopped.
- The obtained value should be put in the table.
- The same procedure is repeated for other squares.
- Then the average values are calculated to obtain the CPI of the sample.

No. of Observations	Courses per Inch (CPI)	Average CPI
1		
2		
3		

5. Determination of Wales per Inch (WPI) from the Given Sample.

- A square with an area of 1 inch^2 (1inch x 1inch) is drawn on the face side of the sample using ballpoint pen with two sides of it being parallel to the wales. Another 2 similar squares are drawn in such a way so that none of the wales or courses of a square is repeated in other squares.
- The number of wales exist between the two sides of the square is counted for all three squares.
- The average is calculated to obtain the WPI of the sample.

No. of Observations	Wales per Inch (WPI)	Average WPI
1		
2		
3		

6.

□ Excellent	[10]
□ Very Good	[9]
□ Good	[8]
□ Satisfactory	[7]
□ Bad	[6]

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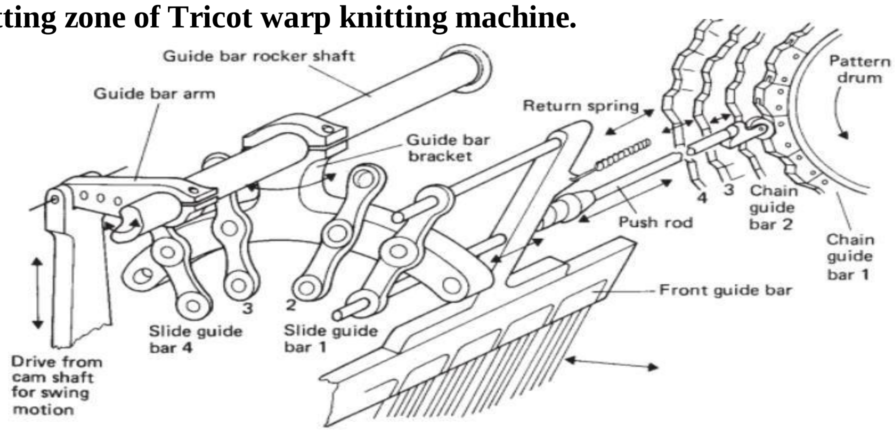
Dept: ____________________Date: _____________

Name: ID:

Exp No: **10** Name of the Experiment: **Study on the Tricot Warp Knitting Machine**

Sign:	Key Words: Mechanism of tricot warp knitting m/c, Swinging & Shogging motion, draw the Lapping diagram. identify the link no, link type, Prepare chain notation.

1. Draw the knitting zone of Tricot warp knitting machine.

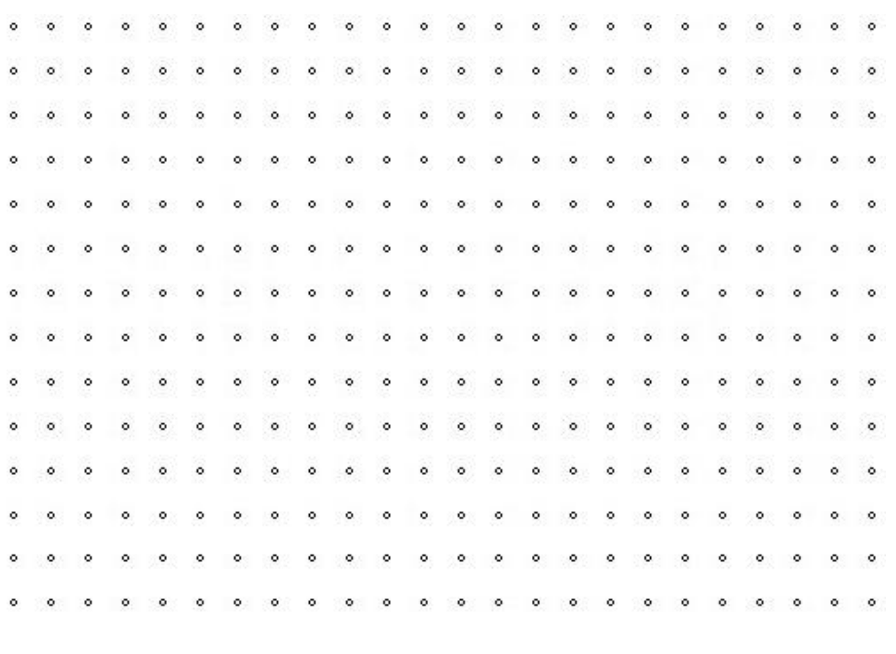


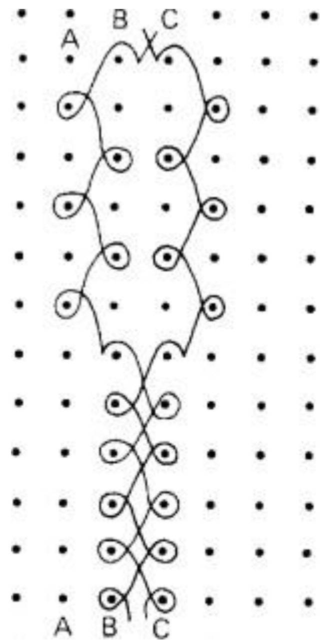
2. List down the differences between weft knitting and warp knitting.

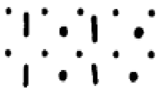
Weft Knitting	Warp Knitting

3. Define Swinging motion and Shogging motion of the guide bar.

4. Analyze the existing fabric and draw its lapping diagram, chain notation and link arrangement.



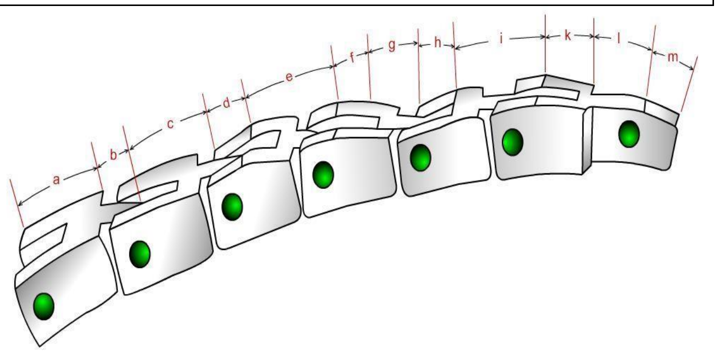


Threading: 

Front Guide Bar	Chain Notation	
	Link Arrangement	
Back Guide Bar	Chain Notation	
	Link Arrangement	

Tricot Chain Link:

These links are designed by letters indicating the ground edge-
'a'- is an underground link (no ground)
'b'- is a link on which the fork is ground
'c'- is a link on which the tail is ground
'd'- is a link on which both the fork and tail are ground



5. List down the 6 basic lapping movements/basic stitches of warp knitting.